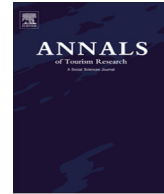




Contents lists available at ScienceDirect

Annals of Tourism Research

journal homepage: www.elsevier.com/locate/atoures



Reject or select: Mapping destination choice



Marion Karl*, Christine Reintinger, Jürgen Schmude

University of Munich, Department of Geography, Luisenstrasse 37, 80333 Munich, Germany

ARTICLE INFO

Article history:

Received 10 June 2014

Revised 21 April 2015

Accepted 15 June 2015

Coordinating Editor: Alain Decrop

Keywords:

Destination choice

Tourist decision-making

Set theory

Tourist typology

Travel constraints

ABSTRACT

The purpose of this study is to empirically explore tourists' destination choice processes. Destination choices are investigated using a combination of data on destinations and on tourists' individual destination choices. Data were collected in Munich/Germany in 2013 using personal interviews; 622 interviews were completed. This approach allows detecting reasons for the rejection or selection of certain types of destinations during the destination choice process. Results show that tourists often start the destination choice process with various combinations of destination types but act similarly when choosing the final destination. The investigation of tourist and destination characteristics results in a tourist typology that varies in regard to similarity and type of alternative destinations at different stages of the destination choice process.

© 2015 Elsevier Ltd. All rights reserved.

Introduction

Choosing a travel destination is a very complex process with many influencing factors. Understanding the underlying destination choice (DC) processes of tourists is a fundamental issue both from an academic and destination management point of view. Most previous studies analysing travel decisions concentrate either on the outcome of DCs following a microeconomic input–output approach (e.g. Papatheodorou, 2001; Seddighi & Theodorou, 2002) or on internal and external influencing factors. These psychological, behaviouristic approaches (structural models: e.g. Um & Crompton, 1990; Woodside & Lysons, 1989; process models: e.g. Mathieson & Wall, 1982; Moutinho, 1987; van Raaij & Francken, 1984) focus on the tourist's behaviour during a travel

* Corresponding author. Tel.: +49 89 2180 4039; fax: +49 89 2180 4099.

E-mail addresses: marion.karl@lmu.de (M. Karl), c.reintinger@lmu.de (C. Reintinger), j.schmude@lmu.de (J. Schmude).

decision-making process. While these models identify cognitive, affective and behavioural influencing factors of DC, characteristics of destination alternatives at the different stages of the DC process and why they are rejected before the final choice remain unexplored.

The aim of this study is to investigate the rejection or selection of destinations during the DC process. To enhance the understanding of the process and structure of DC, this study links destination and tourist characteristics and analyses the interaction between these two aspects. Moreover, this research attempts to shed some light upon tourists' individual DC processes. The purpose is to identify whether and in which way tourists differ in their DC in regard to similarity and type of alternative destinations at the stages of the DC process.

With a long planning period and high personal involvement, DCs are core decisions in the travel decision-making process and on the highest hierarchical order of the three kinds of travel decisions (core, secondary and en route decisions (Fesenmaier & Jeng, 2000)). An approach from consumer research to explain the structure and process of DCs is the set theory (Crompton, 1992; Narayana & Markin, 1975; Um & Crompton, 1990; Woodside & Lysonski, 1989). Set theory is most relevant for purchase decisions that entail intensive information search, evaluation of alternatives, a certain level of perceived risk and high personal involvement (Spiggle & Sewall, 1987), criteria met by DC (Crompton & Ankamah, 1993). According to set theory, DCs are multistage processes where numerous alternative destinations are reduced successively in a funnel-like manner (Sirakaya & Woodside, 2005; Um & Crompton, 1990; Woodside & Lysonski, 1989). Theoretical approaches to set theory focus on the creation of a comprehensive model of DC (e.g. Crompton, 1992; Crompton & Ankamah, 1993), while qualitative studies aim at an in-depth investigation of tourists' decision-making process (e.g. Decrop, 2010). Quantitative studies which apply set theory analyse specific sets of the DC structure (e.g. Um & Crompton, 1992; Woodside & Lysonski, 1989; Woodside & Sherrell, 1977) since it is hardly possible to investigate all DC sets in a quantitative survey. Some subsets are very complex, rather unconscious and difficult to measure. Despite many variations and differences in the terminology, the set model concepts are summarised in a process structure model approach (see Fig. 1).

Alternative destinations are allocated into hierarchically structured sets and only a few destinations are part of the final DC (Crompton, 1992). The allocation in the choice sets depends on internal factors such as socio-psychological characteristics of the tourist, travel constraints as well as external factors related to the distinct features of the destination (Ankomah, Crompton, & Baker, 1996). During the initial phase, prior to the decision to go travelling, several destinations from the total set (i.e. all possible destinations) are grouped in an awareness set (i.e. all known destinations) (Howard & Sheth, 1969). The relevant set (i.e. all considered destinations) is formed once the decision process about taking a trip has been initiated (Um & Crompton, 1990). The total set can be split into three sub-categories: unawareness set (i.e. all destinations a tourist is not aware of (Spiggle & Sewall, 1987)); available awareness set (i.e. all destinations a tourist is aware of and has the ability to visit (Woodside & Sherrell, 1977)); unavailable awareness set (i.e. all known but due to different reasons unavailable destinations (Woodside & Lysonski, 1989)). Narayana and Markin (1975) suggest that the (available) awareness set is further divided into an evoked set (equivalent to the relevant set), an inept set (i.e. destinations which are ruled unacceptable) and an inert set (i.e. destinations which

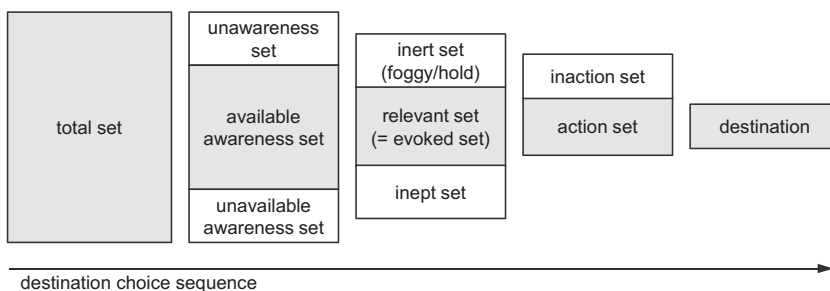


Fig. 1. DC set structure.

are evaluated neither positively nor negatively). Destinations that are evaluated negatively by a tourist are also called excluded set (Decrop, 2006). The last stage of the DC process is the final decision for one destination from the relevant set (Crompton, 1992) which according to Spiggle and Sewall (1987) can be subdivided into an action and inaction set. They proceed on the assumption that the final DC takes place in the action set which encompasses all destinations “towards which a consumer takes some action” (Spiggle & Sewall, 1987, p. 99) to gain more information, for example by contacting a travel agent. Decrop (2010, p. 100) states that in the end of the DC process, a destination is not chosen from the relevant set but from an available set comprising “evoked destinations that really are feasible after considering the vacationer’s constraints”.

Whether or not a destination is able to pass the whole DC process from the awareness set to the final choice depends on a combination of internal and external factors (Crompton, 1992). These factors, inhibitors or facilitators, play an important role in the narrowing-down process of alternative destinations (Crompton & Ankomah, 1993; Um & Crompton, 1992). Studies on inhibitors or travel constraints often investigate tourist characteristics concentrating on a specific market segment (e.g. Gilbert & Hudson, 2000) or a specific destination (e.g. Botha, Crompton, & Kim, 1999; Nyaupane & Andereck, 2007). The focus is on the question whether tourists travel or not, instead of examining the role of constraints in regard to destination characteristics. However, the assumption that restrictive factors in DC can be related to tourist or destination characteristics but are in most cases a combination of both is hardly addressed.

The majority of studies using set theory tends to concentrate on the size and structure of choice sets (e.g. Crompton & Ankomah, 1993; Um & Crompton, 1990; Woodside & Lyonski, 1989; Woodside & Sherrell, 1977) but not on the formation process and the distinct characteristics of the destinations that influence DC. Set models are generally applied as static models although changes in internal or external determinants could lead to displacements within the several sets (Crompton, 1992). The type of alternative destinations changes during the DC process (Decrop, 2010). Nevertheless specific destination attributes which shape the DC process at the different stages are not addressed. Travel decision-making and DC can be seen as an interaction between factors which are more related to the tourist (e.g. travel motives, age) or the destination (e.g. climate, landscape) which has so far been investigated in a more isolated manner in past set theory studies. Moreover, Decrop (2006) as well as Smallman and Moore (2010) criticise that studies on DC are to a great extent theoretical approaches and often lack empirical verification. Some studies on set theory address the relation between sociodemographic characteristics (Crompton & Ankomah, 1993) or travel experience (Woodside & Sherrell, 1977) and set sizes, however, a relation could not be empirically verified yet.

Mapping tourists’ destination choice

Tourist and destination characteristics influence the DC process. This study is driven by four research questions focusing on the interaction between tourist and destination characteristics during the DC process which are explored through a series of descriptive analysis techniques.

Research question 1: What role do tourist and destination characteristics play in the decision to reject a destination during the DC process?

Research question 2: Does the number of alternative destinations at the different stages of the DC process vary according to tourist characteristics?

Research question 3: Are alternative destinations similar within one step of the DC process and different between the steps of the DC process according to destination characteristics?

Research question 4: Do tourist characteristics influence the formation of DC set structures?

Study methods

This study applies a more quantitative approach to gain a broader picture of the variety of destinations that are considered throughout the DC process. A combination of information on tourists’ individual DCs (DC survey) and independent data on destinations (destination index) supports the

detection of reasons for the rejection or selection of certain types of destinations. The examination of the results of the DC structure and process analysis in a more qualitative approach helps to determine and investigate different DC tourist types.

DC survey: sample and design

Data were collected in Munich/Germany in 2013 using a standardised questionnaire in personal interviews. Specific characteristics of the location of data collection (e.g. high proportion of young professionals) are considered throughout the data analysis and interpretation of the results. Trained interviewers approached the respondents in 15 public spaces where people tend to spend time instead of just passing. An initial question ensured that only persons who are planning a main holiday (i.e. at least four overnight stays) within the next twelve months were included in the survey. Only potential tourists at the age of 14 or older were integrated since children influence but are not actively involved in DC (Decrop, 2006). The final sample size is 622. The questionnaire was tested in a two-stage-pretest to check the validity and reliability: First, on a group of 22 university students and faculty members; second, after modification in a survey ($n = 120$) of passers-by.

The questionnaire (three sections, 22 questions) was completed in personal interviews in an average time of 15 min. Details on the next planned trip such as length of stay, holiday type and travel expenses are collected in the first section of the questionnaire. The second section concentrates on general and past travel behaviour and collects data on general motives for travelling and destinations previously visited. In both sections, set theory is applied to analyse DC. Since it is difficult to cover the whole DC set structure with its highly differentiated and complex sets in a quantitative survey, the focus is on a selection of key DC sets which were already successfully investigated in past studies. Participants are asked to name destinations in the following sets: (a) relevant set; (b) action set; (c) initial consideration set; and (d) unavailable consideration set. The relevant set includes alternative destinations which the respondent considers for the next planned holiday and is taken from Crompton (1992) and Ankomah et al. (1996) who suggest that the operationalisation of the late consideration set (here: relevant set) should be associated with a specific time frame. (a) “Which destinations do you consider for your next planned trip (in the following twelve months)?” The action set derives from the relevant set and contains destinations for which the respondent is actively searching for information. (b) “Have you already searched for information on the destination(s) (e.g. travel expenses, sightseeing or accommodation)?” This item is based on Ankomah et al. (1996) advancement of Spiggle and Sewall’s (1987) original study. The initial consideration set contains destinations participants would like to visit in the future and can be seen as a combination of Woodside and Lysonski’s (1989) consideration set (i.e. all destinations considered as holiday destination for a specific time frame) and Crompton’s (1992, p. 422) initial set (i.e. “all the locations that might be considered as potential destinations for a vacation before any decision process about a trip has been activated”). Crompton (1992) modified his concept of the awareness set for a more suitable application in tourism research since it is not the mere awareness of a destination that leads to a visit but the appeal for a destination. In this study, the newly introduced initial consideration set is not limited to a specific time frame to follow this modification but the explicitly expressed wish to visit a destination. (c) “Apart from the next planned trip, which other destinations have you always wanted to visit?” The initial consideration set includes the unavailable consideration set with destinations which are considered but cannot be visited due to constraints. It is partly based on Decrop’s (2010) unavailable set (temporarily unavailable destinations) and expanded since it includes permanent structural (e.g. health, climate) as well as temporal situational (e.g. time, travel companion) constraints. (d) “Are you not able to realise a trip to one or more of these destinations? If yes, please name the reason(s) why the destination(s) is/are not realisable?”

Destination index: data basis and cluster analysis

A destination index is developed to enable the classification and analysis of the numerous destinations mentioned in the survey on a higher level of abstraction using objective data on destinations. The aim of the destination index is to assess tourists’ relative familiarity with a destination and the degree of perceived uncertainty when visiting a new and unknown destination. The concept of familiarity helps to understand DC because tourists rely not only on statistics or objective knowledge when

choosing a holiday destination but also on emotional, affective and tacit knowledge (Williams & Baláž, 2014). Tacit knowledge is lower with unknown destinations and this leads to an increase in uncertainty (Williams & Baláž, 2014) which reflects on DC. Prentice (2006) suggests a framework for an integrated choice set model that includes the concept of affects-as-information, meaning that tourists employ their feelings and emotions as a source of information in DCs. According to this model, familiarity is an important factor that needs to be considered when investigating tourists' decision-making processes. The destination index therefore measures tourists' relative familiarity or novelty of a destination as alluded to by Plog's (1974) psychographic tourist typology of familiarity and novelty seekers. Following this approach, travel behaviour and DC is influenced by certain personality traits such as the level of risk aversion. The psychocentric tourist prefers well-known, highly touristic destinations that offer a certain degree of familiarity. The opposing type, the allocentric tourist, is more likely to travel to destinations that are off the beaten tourist track. The distribution of tourist types changes during a destination's life cycle from a surplus of allocentric tourists in the discovery phase to a surplus of psychocentric tourists at the end of the life cycle (Plog, 1974). This underlines how tourist and destination characteristics are connected and should therefore not be investigated separately.

Based on an extensive literature review on influencing variables in tourists' decision-making as well as expert interviews with practitioners from the tourism industry, the following indicators and indices were chosen to develop the destination index. First, an indicator for tourism intensity (i.e. tourist arrivals per 1,000 inhabitants) (UNWTO, 2013a) which takes into account that a destination has a well-established tourist infrastructure that facilitates travelling and increases familiarity. Another indicator is the relative share of arrivals to a given destination from all German outbound tourists (UNWTO, 2013b). This relies on tourists' awareness of a country as a holiday destination and the possibility to be familiar with a destination either owing to own experience or word-of-mouth information from friends or relatives. Past research shows that both, own experience and word-of-mouth information, have the strongest persuasive power in DCs (Litvin, Goldsmith, & Pan, 2008). Furthermore, the human development index (UNDP, 2013) is added as an indicator for the development of a country as distinguishing mark for infrastructure, public supplies and, probably more important for tourists, healthcare facilities. Accessibility of a destination is chosen as a point of reference for physical distance and is based on a classification of the average flight time from Germany. The greater the distance and the more complicated the access between home and guest country, the lower the level of familiarity with a destination. However, this assumption might not be applicable to all destinations since not only actual geographic but also cognitive distance (i.e. "mental representation of actual distance molded by an individual's social, cultural, and general life experiences" (Crompton & Kim, 2001, p. 512)) is important. Tourists showing a higher involvement with a destination tend to estimate distances to destinations more realistically (Ankomah et al., 1996; Crompton & Kim, 2001). Some countries like the USA are in fact at a great distance to Germany but tourists might nevertheless have a relatively comprehensive knowledge about it and consequently a relatively strong feeling of familiarity. This could be explained by the intensive media representation in the outbound market which may lead to a higher involvement with the destination. Finally, an index for safety and security as a basic prerequisite for a positive tourism development (Reisinger & Mavondo, 2005) is included. This index is composed of seven indicators for organised internal conflict, organised external conflict, political instability, terrorism, homicide, violent demonstrations, and relation with neighbouring countries over a period from 2009 to 2013 derived from the peace index (IEP, 2013).

In order to carry out a cluster analysis, objective data on the destinations from all indicators are classified into five categories (see Table 1). Due to data handling and availability it is necessary to aggregate all destinations on a national scale. The 150 destinations mentioned in the survey are clustered by these five indicators using hierarchical cluster analysis following Ward's minimum variance method with squared Euclidean distance. Squared Euclidean distance is applied as distance measure in this study since similarity is defined by the net distance between values. In order to test the validity of the destination index, a multiple discriminant analysis is conducted. The discriminant functions achieve a high degree of classification accuracy: 94.2% of all cases are classified correctly (see Table 2).

The cluster analysis results in five clusters (see Table 1). The easy travel category (cluster 1) includes very safe and very highly developed destinations at a short distance to Germany with a strong

Table 1

Destinations' categorisation based on cluster analysis.

	Cluster 1 easy travel	Cluster 2 out-of-the-ordinary	Cluster 3 safe adventure	Cluster 4 tricky discovery	Cluster 5 no go
Tourism Intensity ¹ (dominant category)	High 44.4%	Low 60%	Low/medium 44.4/33.3%	Very low 76.7%	Very low 88.9%
German tourism market share ² (dominant category)	Very high 44.1%	Very low/medium–low/low 30/23.3/20%	Medium–low/very low 34.8/30.4%	Very low /low 44.7/23.4%	Very low 66.7%
Human development ³ (dominant category)	Very high 94.4%	High 80%	Very high 72.2%	Medium/low 46.5/39.5%	Low 77.8%
Accessibility/distance ⁴ (dominant category)	Very short/short 47.2/41.7%	Short /medium 40/36.7%	Long/very long 55.6/33.3%	Long 55.8%	Medium-long 55.6%
Safety & security ⁵ (dominant category)	Very high 75%	Medium/low 50/30%	Very high /high 50/33.3%	Low 53.5%	Very low 100%
Destination (exemplary)	Germany, France, Italy, Croatia	Albania, Macedonia, Oman	USA, Canada, Australia, New Zealand	Myanmar, Vietnam, Senegal	Iraq, Pakistan, Sudan, Syria

¹ Tourist arrivals per 1,000 inhabitants: very high (>2000), high (1000 ≤ 2000), medium (500 ≤ 1000), low (100 ≤ 500), very low (<100).

² Classified numbers of arrivals of German tourists: very high (>1%), high (0.5 ≤ 1%), high-medium (0.1 ≤ 0.5%), medium–low (0.05 ≤ 0.1%), low (0.02 ≤ 0.05 %), very low (<0.02%).

³ HDI Rank: very high human development, high human development, medium human development, low human development, very low human development.

⁴ Average flight time from German airport: very short (<2 h flight), short (2 ≤ 4 h flight), short-medium (4 ≤ 6 h flight), medium-long (6 ≤ 10 h flight), long (10 ≤ 15 h flight), very long (> 15 h flight).

⁵ Aggregated indicators derived from Peace Index: very safe, safe, medium, unsafe, very unsafe.

Table 2

Classification results of the multiple discriminant analysis.

Actual group	Predicted group membership					N
	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	
Cluster 1	35	0	1	0	0	36
	97.2%	0.0%	2.8%	0.0%	0.0%	
Cluster 2	1	29	0	0	0	30
	3.3%	96.7%	0.0%	0.0%	0.0%	
Cluster 3	0	0	18	0	0	18
	0.0%	0.0%	100.0%	0.0%	0.0%	
Cluster 4	0	2	1	37	3	43
	0.0%	4.7%	2.3%	86.0%	7.0%	
Cluster 5	0	0	0	0	10	10
	0.0%	0.0%	0.0%	0.0%	100.0%	

Percent of grouped cases correctly classified: 94.2%.

tourist flow from Germany. Destinations from cluster 1 are mostly located in Western and Central Europe. Out-of-the-ordinary destinations (cluster 2) are mostly highly developed, at medium distance but with a low general and German market tourism intensity. These destinations are spatially more disperse and located in Eastern Europe, North Africa and the Middle East. The safe adventure category (cluster 3) is characterised by very safe, very highly developed destinations at a long distance to

Germany. Safe adventure destinations can be found in North America, Australia and partly in South America (Argentina, Chile). Tricky discovery destinations (cluster 4) are also at a great distance to Germany but are, in contrast with cluster 3, rather unsafe, non-touristic and less developed. Most destinations in South and Central America are defined as tricky discoveries. Other destinations from this cluster are placed in Sub-Saharan Africa and South/South East Asia. Destinations in the no go cluster (cluster 5) are defined as very unsafe and undeveloped with hardly any tourism and located in countries in Africa and Asia that are affected by severe (military) conflicts.

Results & discussion

The result of the cluster analysis, represented in the destination index as a categorisation of all destinations according to the level of familiarity, is used to analyse and structure the individual DC processes of German tourists. Destinations mentioned in the survey are grouped according to the destination index, which results in a dataset containing the respondents' DC sets linked with information concerning the destinations' level of familiarity. This dataset is used for further statistical analyses to investigate DC more closely.

Travel constraints

The relative strength of beliefs about facilitators and inhibitors of destinations is an important factor in the selection or rejection of a destination (Um & Crompton, 1992). Travel constraints which are related to the destination are likely to vary according to the type of destination (represented by the cluster category of the destination index). Nevertheless, tourists perceive constraints differently depending on individual characteristics (Nyaupane & Andereck, 2007). Travel constraints should therefore vary depending on the destination and the tourist.

To investigate the first research question respondents are asked to name reasons why an initially considered destination can currently not be visited (i.e. measured on a nominal scale with predefined answers and the possibility to expand the item list). In this survey travel constraints are perceived constraints and may not reflect the actual situation in some cases as it often happens with risk perceptions (Sönmez & Graefe, 1998). The main constraints for initially considered destinations are financial and time restrictions (see Table 3). This complies with Botha et al. (1999) who show that financial reasons are amongst the main constraints for visiting an ideal dream destination. The factors family situation, dangers at the destination (e.g. natural disasters, terrorism or criminality), lack of adequate travel companion as well as political situation (e.g. political instability, violent demonstrations, disapproval of the government), health constraints and unfavourable climate conditions are gradually named less frequently. These constraints are hence less relevant at this stage of the DC process. Since safety and security aspects are often seen as very important factors in DC (Sönmez & Graefe, 1998), it seems possible that destinations with these constraints have already been rejected before this DC stage.

Table 3
Relationship between the variables cluster allocation and travel constraints.

	N	Sig.	df	χ^2	Cramér's V	Cluster 1 (in percent)	Cluster 2 (in percent)	Cluster 3 (in percent)	Cluster 4 (in percent)	Cluster 5 (in percent)
Financial constraints	535	0.004**	4	15.372	0.131	54.1	53.6	63.8	61.7	11.1
Time constraints	372	0.009**	4	13.554	0.123	47.6	45.2	37.0	44.5	0.0
Family situation	101	0.164	4	6.506	0.085	–	–	–	–	–
Dangers at the destination	56	0.000***	4	112.612	0.356	0.0	13.4	2.7	10.6	77.8
Travel companion	54	0.001**	4	18.858	0.145	12.9	6.0	4.9	3.4	0.0
Health constraints	29	0.092	4	7.983	0.092	–	–	–	–	–
Political situation	25	0.000***	4	53.342	0.253	0.6	12.3	0.0	5.3	25.0
Climate conditions	22	0.204	4	5.934	0.204	–	–	–	–	–

** $p < 0.01$.
*** $p < 0.001$.

Pearson's chi-squared test is used to examine the relationship between the type of destination and travel constraints to determine if certain destinations are rejected due to certain restrictions (see Table 3).

Some constraints vary according to the type of destination that tourists would like to visit and are probably more related to the destination than to the tourist. Financial constraints, lack of holidays, impossibility to find an adequate travel companion, dangers and political situation in a destination are statistically dependent on the type of destination. Financial constraints dominate in destinations from cluster 3 and 4 and apply less to destinations from cluster 1 and 2. This can be explained by the geographic location of these destinations since travel distance and mode of travel are significant influencing factors of German tourists' DCs (Marcussen, 2011). Destinations from cluster 3 and 4 require in most cases long-distance journeys associated with high travel expenses. A bias in the estimation of travel costs may exist which can be explained by an overestimation of distances to relatively unfamiliar destinations (Ankomah et al., 1996; Crompton & Kim, 2001) and an unrealistic assessment of on-site travel costs at unfamiliar destinations due to lack of knowledge.

Respondents who reject a destination from cluster 5 for safety and security reasons (represented by the items "dangers at the destination" or "political situation") do not consider other constraints because these constraints are more distinct than others. Differences between the five clusters in regard to time constraints and lack of travel companion are comparatively small. These constraints are probably more related to tourist than destination characteristics or it is an interaction between both characteristics. The age group 20–29 years can help to explain this statement. Here, the most dominant travel constraints are financial reasons and these tourists are particularly interested in destinations from clusters 3 and 4 which are significantly linked to financial constraints. Consequently, tourist (e.g. young age, low income and preference for long haul holidays) and destination (e.g. destinations with high (perceived) travel costs) characteristics play an interacting role between travel constraint and DC.

Neither tourist nor destination related constraints are restricted to temporal situational (e.g. one time natural disaster; young children in the family) or permanent structural (e.g. unfavourable climate conditions; chronic diseases) constraints. In fact, tourists' DC structures may change over time as tourists' climb the travel career ladder (Ryan, 1998) or changes in the actual situation at a destination may occur.

Size of destination choice sets

Travel constraints act as filters that help to successively reduce alternative destinations during the DC process in a funnel-like manner. In this study, 95% of all respondents have no more than five destinations in the initial consideration set, three destinations in the relevant set and two destinations in the action set. This corroborates findings of a great deal of the previous work on DC sets and supports the thesis that DC follows a funnel-like process (Crompton, 1992; Crompton & Ankomah, 1993; Jeng & Fesenmaier, 2002; Moutinho, 1987; Sirakaya & Woodside, 2005) and that the relevant set is composed of no more than a few destinations (Crompton & Ankomah, 1993; Decrop, 2010; Um & Crompton, 1990; Woodside & Lysonski, 1989). Due to differences in the methodology and data collection the results of the average set sizes often vary. This underlines the high sensitivity of data collection procedures as well as wording of questionnaires in this research area and limits the comparability of this study's results to previous set theory studies (e.g. Crompton & Ankomah, 1993; Woodside & Lysonski, 1989; Woodside & Sherrell, 1977), at least in regard to the average set sizes. An analysis on set size variation during the DC sequence in this study reveals that the average set size of the relevant set

Table 4
Descriptive statistics of DC set sizes.

	M (SD)	Mode	Min	Max
Initial consideration set	2.31 (1.67)	1	0	10
Unavailable consideration set	1.72 (1.54)	1	0	10
Relevant set	1.93 (0.94)	1	0	5
Action set	0.68 (0.90)	0	0	5

is 1.9 and of the initial consideration set is 2.3 (see Table 4). The ratio of 0.8 between relevant and initial consideration set size is in the range identified by Crompton and Ankamah (1993).

A further analysis of respondents' DC set sizes related to research question 2 reveals that the number of alternative destinations for a specific trip (relevant set size) is not dependent on tourist characteristics, while the number of initially considered destinations at an earlier (hypothetical) stage of the DC is correlated with these factors. One explanatory approach is the concept of choice overload which suggests that too many choice alternatives can have a negative influence on decision-making (e.g. making no choice) (Park & Jang, 2013). This is particularly relevant in tourism since "the number of products approaches infinity" (Park & Jang, 2013, p. 2) due to multiple options concerning aspects such as accommodation or travel modes. Tourists may react to choice overload by only considering very few destinations at the final choice moment. This does not apply to initially considered destinations since there is no need for the tourist to actively choose between alternatives yet.

Since homogeneity of variances as a precondition for one-way analysis of variance (ANOVA) is not given, Kruskal–Wallis one-way analysis of variance as a non-parametric method is applied to explore differences in DC set size and sociodemographic as well as travel-related variables. The sizes of the relevant and action sets do not differ significantly amongst the age cohorts and educational level groups. Nevertheless, a tendency for the relevant set can be identified, which partly supports Crompton and Ankamah (1993) who state that a positive correlation between relevant set size and educational level exists. However, the size of the initial and unavailable consideration sets varies significantly between groups of educational levels, age classes and travel frequency (i.e. number of main holidays in the last three years; see Table 5).

A higher level of education corresponds with a higher number of alternative destinations in the initial consideration set. Tourists with a higher educational level are significantly more interested in cultural/educational than in sun-sand-sea holidays. van Raaij and Francken (1984) state that persons with a high educational level show more interest in the destination than other tourists. Consequently, the destination itself plays a more important role in DC for these tourists, while other tourists might focus on different aspects (e.g. sunshine guarantee). The initial consideration set of respondents with the highest educational level however is smaller than the initial consideration set of the second highest education group. This might be due to the fact that the latter consists to a large extent of young university students who have the largest initial consideration set size since age and initial consideration set size are highly correlated. The number of alternative destinations in this set decreases with an increasing age, except for the youngest age group. Respondents from the age 14 to 19 years have comparatively small initial consideration sets, probably because they are at the beginning of their travel career or are not the main decision makers if they travel with their parents. Tourists in the age group from 30 to 59 years are often restricted in their DC by their family situation. It is unclear why the initial consideration set of elderly tourists is the smallest. Two reasons may apply. First, older people are more realistic in their DCs and consider fewer destinations to be visited in the near future with a higher probability. Second, older tourists have already seen many of their initially considered destinations which is why the initial consideration set size has decreased gradually during their life. Similar observations result from an analysis of the unavailable consideration set, except for the group with the highest educational level. This group has a large initial but small unavailable consideration set. A reason might be the higher average income of this group in comparison to the second highest educational level, which facilitates the realisation of travel plans.

The influence of travel frequency on set size is partly accepted. Set sizes in the earlier stages of the DC process vary significantly according to the level of travel frequency, while this does not apply to set sizes of the later stages. Frequent travellers (i.e. at least three main holidays in the last three years) consider significantly more destinations throughout the DC process than interval (i.e. one or two main holidays in the last three years) and infrequent travellers (i.e. no main holiday in the last three years) who have the lowest number of destinations in their initial and unavailable consideration set. The great importance of the travel motives "culture/education" ($p = 0.047$) and "gain new impressions from another country or society" ($p = 0.000$) for frequent travellers indicates that the actual destination of a holiday plays a more important role in their DCs.

Table 5

Relationship between initial/unavailable consideration set size and tourist characteristics.

Test variable	Grouping variable	Mean (SD)	Kruskal–Wallis H	df	Sig.
Initial consideration set size	Educational level ¹		36.454	4	0.000***
	Lower secondary education	1.60 (1.38)			
	Secondary education	1.74 (1.12)			
	Advanced secondary education	2.72 (1.81)			
	University degree	2.33 (1.69)			
	Age		80.014	6	0.000***
	14–19 years	1.96 (1.51)			
	20–29 years	2.94 (1.80)			
	30–39 years	2.58 (1.60)			
	40–49 years	2.10 (1.37)			
	50–59 years	1.66 (1.58)			
	60–69 years	1.63 (1.19)			
	70 years and older	1.36 (1.19)			
	Travel frequency		44.456	2	0.000***
Unavailable consideration set size	Frequent traveller	2.82 (1.84)			
	Interval traveller	1.91 (1.40)			
	Infrequent traveller	1.76 (1.54)			
	Educational level ¹		21.390	4	0.000***
	Lower secondary education	1.69 (1.42)			
	Secondary education	1.19 (1.11)			
	Advanced secondary education	2.01 (1.62)			
	University degree	1.64 (1.60)			
	Age		42.497	6	0.000***
	14–19 years	1.90 (1.57)			
	20–29 years	2.03 (1.51)			
	30–39 years	1.95 (1.68)			
	40–49 years	1.57 (1.38)			
	50–59 years	1.38 (1.72)			
	60–69 years	1.04 (1.30)			
	70 years and older	0.88 (0.83)			
	Travel frequency		15.220	2	0.000***
	Frequent traveller	2.00 (1.66)			
	Interval traveller	1.47 (1.38)			
	Infrequent traveller	1.70 (1.49)			

Note: Only significant factors are listed in this table.

*** $p < 0.001$.

¹ Others ($n = 21$).

Structure of destination choice processes

The investigation of the cluster composition concerning research question 3 detects remarkable changes during the DC sequence. Two important shifts are identified: A minor shift in cluster composition between initial and unavailable consideration set and a major shift from initial consideration to relevant set (see Table 6).

In the minor shift from initial to unavailable consideration set, the share of cluster 3 and 4 combined increases. This is due to travel constraints which deter a tourist from visiting a destination of these two clusters at the moment (see Table 3). Moreover, the share of cluster 1 is reduced considerably from initial to unavailable consideration set. The observed decrease could be attributed to the characteristics of destinations in this cluster being mostly safe, close to the home country, with a highly developed (touristic) infrastructure and are therefore easy travel destinations. In the second shift from initial consideration to relevant set, the share of cluster 1 increases noticeably, while the share of cluster 3 and 4 declines. The same reasons as for the changes from initial to unavailable

Table 6

Cluster composition in DC sequence.

	N	Cluster 1 (in percent)	Cluster 2 (in percent)	Cluster 3 (in percent)	Cluster 4 (in percent)	Cluster 5 (in percent)
Initial consideration set	1342	27.0	9.3	35.9	27.1	0.7
Unavailable consideration set	903	18.9	9.4	40.6	30.1	1.0
Relevant set	1208	72.5	8.9	11.1	7.5	0.0
Action set	729	72.8	8.0	11.1	8.1	0.0
Destination choice	479	72.0	10.2	11.1	6.7	0.0
Past destination choice	1319	73.5	9.2	8.9	8.4	0.0

consideration set may apply here. Cluster 2 remains stable during the whole DC sequence. It contains many destinations that Germans with a migrant background frequently visit. Visiting family and relatives is the most commonly mentioned decisive motive for travelling to cluster 2. Destination characteristics such as the level of familiarity are probably less relevant for tourists whose main travel motive is to see their relatives or friends. In this case, no real alternative destination exists and the final destination is therefore less variable compared to tourists with other travel motives. In this sample destinations from cluster 5 only occur in the initial and unavailable consideration set. Political instability and terrorism have a strong negative influence on DC (Sönmez & Graefe, 1998), which in regard to the most important travel constraints for destinations from cluster 5, could explain the absence in the relevant set and past DCs. No significant change is detected in the sequence from relevant to action set and to the final choice of one destination for the planned trip. As can be seen from Table 6, the cluster composition of the more realistic DC sets (i.e. relevant and action set) as well as the final DC match past travel decisions. This supports Decrop (2010) who states that many destinations in the relevant set can be deduced from past travel experiences, in many cases even from the last trip. Moreover, post-trip evaluation is an important factor in DC as it helps to support future DC (Martin & Woodside, 2012).

The similarity within DC sets is tested on the homogeneity of the sets. In order to measure homogeneity of cluster allocation in the DC sets, an Index of Destination Differentiation (IDD) is developed. The IDD is calculated analogous to the Index of Economic Differentiation (Amemiya, 1963) which estimates the level of differentiation of economic sectors. In this study the IDD estimates the level of dissimilarity of alternative destinations at the key stages of the DC process. The IDD enables the comparison of choice sets in regard to variety in order to draw conclusions about alternative (competing) destinations in the respective choice set. Values between 0 (even distribution to all clusters) and 1 (concentration on one cluster) can be assumed. The larger the number of clusters (n) in a set, the smaller the IDD and the more heterogeneous the choice set.

$$IDD_{\text{choice set}} = \sum_{i=1}^n \frac{n}{n-1} \left(p_i - \frac{1}{n} \right)^2$$

IDD: Index of Destination Differentiation; $0 \leq IDD \leq 1$

p_i : proportion of cluster i in choice set; $0 \leq p_i \leq 1$

n : number of clusters in choice set.

The IDD is calculated on an aggregated level for each DC set. The initial and the unavailable consideration sets are the most differentiated and heterogeneous choice sets with an IDD of 0.105 and 0.128, respectively. Later sets in the DC process are more homogeneous with a stronger concentration on one type of destination: $IDD_{\text{relevant set}} = 0.393$; $IDD_{\text{action set}} = 0.410$. Past DCs are the least differentiated ($IDD_{\text{past DC set}} = 0.427$).

The results of the analysis of cluster composition and homogeneity in the DC sets reveal that while most respondents consider a greater variety of destinations from all cluster categories in their initial consideration set, they mostly choose their potential destinations for a specific trip within one cluster category and travel to one cluster category. DC becomes more realistic by the end of the DC process

(Decrop, 2010) and consequently familiar destinations from cluster 1 which can be visited more easily dominate in the homogeneous sets by the end of the DC process. This implies that competitive destinations for one distinct trip are similar in regard to familiarity. Although travel decisions are associated with a high level of involvement (Moutinho, 1987), habitual or rational behaviour also shapes travel decision-making leading to revisit familiar destinations or cautious DCs (Decrop & Zidda, 2006). Two factors have to be taken into account here: information search costs and risk reduction tendency. Tourists' motivation to deal with different alternatives and gather information to make DCs are limited since every information search is bound by costs (time or money) (Mansfeld, 1992). When tourists decide to visit familiar destinations, probably even already visited destinations, they are able to reduce information search costs. Furthermore, "the unique characteristics of tourism products can cause unexpected outcomes compared with retail products" (Park & Jang, 2013, p. 2). If tourists concentrate on familiar destinations, they are able to better estimate the outcome of their DC and reduce risks for their holiday. Facilitators are more meaningful at the beginning of the DC (Um & Crompton, 1992) and assist the inclusion of those destinations into the initial consideration set which would lead to an ideal DC outcome. In contrast, inhibitors and potential risks play a more important role at the end of the DC process and a destination is chosen which offers "a satisfactory pay-off which can be attained within perceived constraints" (Um & Crompton, 1992, p. 24).

Tourist typology of destination choice

A tourist typology of DC structures can help to explain the interaction between destination and tourist characteristics in the DC process. In contrast with previous studies on travel decision-making typologies which concentrate on various factors concerning the tourist such as personality traits (e.g. Cohen, 1972; Plog, 1974), this study focuses on the key points of the DC structure.

Based on the results of the DC structure analysis, set structures are examined in more detail on the level of the individual tourist. The tourist typology is developed using a combination of classification techniques and qualitative data mining. The qualitative data mining (i.e. manual extraction of patterns from the data set) applied here as statistical cluster analysis is not possible for this data set due to the large number of different variable values. In the first step, a DC set type is developed for each set that integrates all mentioned destination clusters for one respondent. Subsequently, all respondents with the same DC set structure (composed of DC set types in the relevant, initial consideration and past DC set) are grouped in one tourist type. The most common and distinctive groups are further analysed and represented by case examples. These case examples are applied to study the fourth research question whether tourist characteristics such as sociodemographic and travel-related factors are decisive determinants in the formation of DC structures.

As previous studies on set theory as well as statistical tests of this study show, set sizes vary depending on tourist characteristics. One-way analysis of variance (ANOVA) is calculated in order to detect differences in the initial and unavailable consideration set sizes between the DC tourist types (see Table 7). Pearson's chi-squared test shows that the factors age, gender, travel frequency and travel motives (discovery/extraordinary experience, culture and education, risky experience, time with family/friends) are significant determinants of DC set structures.

The down-to-earth tourist types (realistic planner, safe dreamer) are characterised by a realistic DC set structure with safe and familiar destinations. Realistic planners such as respondent 179, who is female, 40–49 years old and highly educated, have a high congruency in their imagination of travelling, travel wishes and actual travel behaviour. In this case, alternative destinations for her next trip are Germany and France, she would like to visit Ireland and Portugal in the future (none of them are perceived to be unavailable) and she has already visited the destination which she considers for her next trip. Like the average down-to-earth tourist types, she is a less frequent traveller with small initial and unavailable consideration set sizes. This respondent prefers individually organised holidays and is particularly interested in cultural/educational or natural themes as well as the country and people of the visited destination. Safe dreamers on the other hand are less interested in culture and education while travelling. This travel motive is highly linked to the variable age which is one of the most important distinctive factors between these two tourist types (i.e. realistic planners are generally older than safe dreamers). The selected exemplary respondent 565 for safe dreamers is female, between 14 and 19 years old and has not (yet) graduated. The planned nature/camping trip with her family to

Table 7

Distinct characteristics of the DC tourist typology.

	Sig.	Realistic Planner		Safe Dreamer		Unrealistic Planner		Day Dreamer		Adventurer	
Set composition											
Relevant set		Cluster 1		Cluster 1		Cluster 1–4		Cluster 1		Cluster 1–4	
Initial consideration set		Cluster 1		Cluster 1/3		Cluster 1–5		Cluster 2–5		Cluster 1–5	
Past destination choice		Cluster 1		Cluster 1		Cluster 1		Cluster 1		Cluster 1–5	
Set size (mean)		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Initial consideration set	0.010 [*]	1.9	1.0	2.0	1.2	2.8	1.5	3.2	1.7	2.8	1.5
Unavailable consideration set	0.000 ^{***}	0.9	1.0	1.5	1.2	1.5	1.2	2.5	1.8	2.0	1.5
Age	0.000 ^{***}										
Overrepresented group		40–49 years 60–69 years 70 years or older		14–19 years		20–29 years		30–39 years		20–29 years 50–59 years	
Underrepresented group		20–39 years				60–69 years		60–69 years 70 years or older		14–19 years 40–49 years 70 years or older	
Gender (in percent)	0.002 ^{**}										
Female (55.3)		59.6		43.3		48.5		71.4		34.1	
Male (44.7)		40.4		56.6		51.5		28.6		65.9	
Travel regularity (in percent)	0.014 [*]										
Regular traveller in past 3 years (44.0)		30.8		34.2		40.0		52.6		65.0	
Travel motives (in percent)											
Discovery, extraordinary experience (35.2)	0.031 [*]	24.5		32.9		48.6		37.2		60.0	
Culture and education (46.5)	0.019 [*]	54.7		27.6		45.7		48.7		50.0	
Risky experience (13.7)	0.000 ^{***}	3.8		13.2		5.7		11.5		45.0	
Time with family/friends (42.3)	0.016 [*]	50.9		43.4		40.0		64.1		30.0	

Note: Only significant factors of tourist destination choice typology are listed in this table.

^{*} $p < 0.05$.

^{**} $p < 0.01$.

^{***} $p < 0.001$.

either England or France (which she has visited in the past) is a typical holiday for her. Destinations in her initial/unavailable consideration set (Australia, USA) reflect the distinguishing feature between the down-to-earth tourist types. Realistic planners initially consider only safe and short-distance destinations (cluster 1), safe dreamers, however, also safe and long-distance destinations (cluster 3). The respondent's travel motives (relaxing, discovering, spending time with family/friends) and the relative low cultural interest while travelling are in line with the average safe dreamer.

In contrast with down-to-earth tourists, more unrealistic tourist types indicate a rather imaginative DC with initial consideration set sizes above average. A typical unrealistic planner is respondent 75. She is 20–29 years old, has a high educational level and her initial consideration set (China, Chile/Argentina, South Africa, Vietnam, Laos, Cambodia, Morocco) as well as her relevant set (Canada, India) are composed of destinations from various clusters except for cluster 1. Furthermore with France, Germany and England, she has only travelled to destinations from cluster 1 in the past three years. Like many unrealistic planners, she prefers extraordinary but non-risky relaxing holidays. As opposed to unrealistic planners, day dreamers have a more realistic DC structure and do not consider destinations in the relevant set which cannot be realistically realised in the near future. An example for the day dreamer tourist is respondent 299, a female tourist between 20 and 29 years with a high educational level. Her plan is to go on a package trip and to spend a relaxing but sporty nature/sun holiday with her family or friends in Austria, Croatia or Italy, destinations which she has visited before. Nevertheless, she would like to visit USA, Brazil, India, China or Thailand and does not judge them as unavailable for future holidays. A remarkable characteristic of day dreamers is the high travel frequency. Apparently travelling is important to day dreamers even though they are not able to realise initially considered destinations at the moment. Day dreamers do not postpone all travel plans but substitute travel destinations which are not realisable at the moment with other destinations. An explanation can be the current family situation as a possible constraint that strongly influences DCs. The high relative importance of the travel motive "spending time with family/partner/friends" in this tourist type and sociodemographic factors (higher proportion of female tourists and the age group 30–39) support this argument.

The least familiarity-seeking adventurer tourist type has a fairly realistic DC set structure like respondent 472, a male tourist between 20 and 29 years with a high educational level. His relevant set (China, Australia), past DCs (China, Peru, Bolivia) and initial consideration set (Israel, Africa, Eastern Europe) are all composed of destinations from various cluster categories except for cluster 1. Thus, actual and imagination of travel behaviour are highly consistent. This frequent traveller is planning an individually organised sport/cycling/hiking trip. Like many adventurer tourists, this respondent has a different motivation to go travelling than other tourist types. He is attracted by extraordinary experiences and cultural/natural aspects while travelling but less motivated to go on a holiday in order to spend time with family and/or friends. None of the initially considered destinations are perceived to be unavailable. Due to the dominant age groups (20–29 and 50–59 years), adventurer tourists might be less restricted by familial reasons, have a certain level of travel experience and are therefore able to carry out travel plans more easily.

Conclusion

Tourists' DCs are complex processes with various determinants that decide whether or not a destination moves from one step of the DC sequence to the next and is finally chosen as a travel destination. Set theory helps to reduce some of the complexity and allows investigating several alternative destinations during the choice process even if these destinations are finally not chosen.

Theoretical contribution

This empirical study enhances the understanding and research methodology of the DC process by applying set theory which was often used as a conceptual model lacking empirical evidence (Decrop, 2010). In particular, this research determines whether there is a link between the stage in the DC process and the kind of destination that is considered at that stage. The construction of a destination index in this study based on objective destination characteristics instead of tourists' subjective estimation like the "individualised index of attraction" (Deutsch-Burgner, Ravualaparthi, & Goulias, 2014) allows to analyse the influence of destination attributes on the selection or rejection of a destination.

A significant finding to emerge from this study is that DC sets vary in regard to variety and type of dominant destinations. The initial consideration set consists of a wide range of destinations from various clusters that is narrowed down during the DC process and the definite travel behaviour. The more realistic DC sets are dominated by close, well-known and safe destinations with a highly developed tourist infrastructure. This supports Plog's theory (1974) as it emphasises the importance of familiarity in DC. Moreover, this study is able to identify a tourist typology of DC that is characterised by similar DC structures. Decrop and Snelders' (2005) or Decrop and Zidda's (2006) tourist typology (based on decision-making behaviour) is partly reflected in these tourist types. For example, rational vacationers with a high risk avoiding tendency, rational decision-making behaviour and realistic DC structure are comparable to realistic planners in this study. Furthermore, hedonistic vacationers are similar to day dreamers since these tourists also tend to plan their dream holiday without consideration of travel constraints. This study can be seen as an enhancement of past tourist typologies since it includes the specific characteristics of the destinations as an additional explanatory factor for the formation of DC tourist types.

The results of this study underline that not only final DCs are essential in the investigation of travel decisions. Tourists often start the DC process with various combinations of destination types but decide very similarly when it comes to choosing the final destination. This specifies Decrop's (2010) findings that the degree of realism increases during the DC process: "Vacationers may move from a preference/ideal value level [...] to an expectation level [...] and finally to a tolerance level [...] as far as plans evolve" (Decrop, 2010, pp. , 110–111). The present study deepens the knowledge about changes in the DC process as it furthermore includes information on the destinations at the different steps of the DC process. One of the few approaches including destination characteristics is Marcussen's (2011) microeconomic study that analyses several influencing factors on DC. However, this study concentrates on the outcome of DC and is therefore not able to answer the question why destinations are

rejected during the DC process. This study shows that besides tourist characteristics, destination related constraints also play an important role in the selection or rejection of destinations during the DC process. Another contribution of this study is the integration of travel constraints into DC, an important factor which has not been considered in many set theory studies. This helps to understand why certain destinations are not realistically considered but excluded for the present time.

Managerial contribution

Travel constraints which occur by the end of the DC process are often more related to personal inhibitors and consequently marketing strategies might be less efficient here (Crompton, 1992). In contrast with this, a strategic marketing which helps to change the perception of travel constraints concerning destinations at the beginning of the DC process (e.g. safety and security image, pricing strategy) would prove to be more successful. For example, the island of Crete has dealt with the discrepancy between the actual and perceived safety and security situation in their destination in a marketing campaign which directly addresses this problematic. Their video campaign shows a tourist who obviously enjoys a peaceful holiday at the destination while frequently observing media reports from his home country that portray a situation of a chaotic destination full of protest and crisis. Destinations that have to deal with a discrepancy between perceived and actual travel costs could apply a similar strategy. Many long-distance destinations are perceived to be unavailable to the tourist due to financial reasons. Nevertheless, the cumulative costs of the whole holiday may in fact be less in comparison to short distance destinations because of lower on-site expenses despite relatively high journey costs.

More detailed information on travel constraints may also be of specific interest to destination management and tourism organisations. Destinations managers with more knowledge about tourists who initially consider their destinations but rate them as unavailable for the moment are able to develop a more target-group-specific marketing strategy. They can address their marketing to tourists who would initially be interested in the destination but decide to travel to another destination in the end. Another aspect which might be of relevance in this context is the identification of competing destinations for one specific trip but also for future holidays (i.e. all destinations in one tourist's relevant or initial consideration set). If destination managers know competing destinations, amenities and the unique selling proposition of a destination can be addressed specifically in comparison to competing destinations.

Limitations and further research

One limitation of this study is the transferability to other source markets and the generalisability of statistical research findings. Cultural factors influence tourists' decision-making processes in several aspects such as information search or travel purchase behaviour (Money & Crotts, 2003). Beside geographic distance which leads to an increase in dis familiarity, cultural distance (i.e. "the degree to which cultural values in one country are different from those in another country" (Sousa & Bradley, 2006, p. 52)) should also be considered. A great difference between culture or habits of home and host country producing a feeling of uncertainty concerning one self's behaviour (Lepp & Gibson, 2008), most likely reinforces the feeling of dis familiarity towards a destination. The high uncertainty avoidance of German tourists (Hofstede, Hofstede, & Minkov, 2010) should be borne in mind as it leads to a tendency towards safe decisions with a low level of uncertainty which might explain the dominance of relatively easy travel destinations in the relevant set and past DCs. The study's findings should thus be considered in regard to specific features of the German source market such as the very high and stable travel intensity or the strong outbound market (Lohmann & Aderhold, 2009) which lead to a high (international) travel experience. A similar study in a country with a different tourism market structure (e.g. weak outbound market) and lower uncertainty avoidance (Hofstede et al., 2010) such as the USA would help to clarify the influence of specific market structure and culture. Although the DC tourist types are case examples for the German source market and do not allow a generalisation of the findings, this exploratory approach nevertheless indicates that certain patterns of DC structures exist.

Several research questions concerning tourists' DC set structures remain unanswered at present. DC is in most cases a joint decision between travel partners (van Raaij & Francken, 1984) and cannot be investigated without consideration of the social environment (Mansfeld, 1992). Therefore, destinations are not only located in tourists' relevant sets because of personal preferences but because of mutual decisions with travel companions (Jang, Lee, Lee, & Hong, 2007).

Moreover, DC is not a static but dynamic concept that is exposed to external and internal influencing factors (Crompton, 1992; Decrop, 2006; Decrop, 2010; Um & Crompton, 1990). The application of a quantitative approach focusing on the key decision points does not allow investigating the formation of DC sets in this dynamic process. A more qualitative and longitudinal approach, as Decrop (2010) shows, will help to shed light on the role of influencing factors in the formation of DC sets. Qualitative methods are furthermore able to detect the unconscious elements of DC (Martin & Woodside, 2012). Moreover, the second step of the DC process—the evaluation of alternative destinations in the latest set and the selection of one destination (Um & Crompton, 1990)—is not investigated in this study. Factors influencing the rejection of a destination in the early stages of the DC process are probably of another nature than factors which lead to the selection of one final holiday destination (Perdue & Meng, 2006) and should therefore be included in future studies.

DC is not concentrated on one but often on several consecutive trips for some years. Destinations which are excluded for one holiday do not necessarily leave the DC structure but can be postponed for the next DC (Decrop, 2010). An important trend in German tourism is the increase in the number of holidays per year (Lohmann & Aderhold, 2009) which leads to an intensifying complexity of travel decisions. A DC is now simultaneously dealing with several trips. This suggests that tourists' DCs not only concern several consecutive trips which are combined in macro-evoked sets (i.e. "set of product alternatives which are all possible in the near future" (Decrop, 2010, p. 102)) but also several types of holidays. Some destinations will shift between holiday DCs but other destinations which are strongly linked to a certain type of holiday will only be relevant for decisions concerning this type of holiday. It seems therefore possible that several parallel relevant sets exist as a sub-category of the macro-evoked set. Woodside and Sherrell (1977) as well as Decrop (2010) state that mode of travel or length of stay, respectively, influence DC sets. Other important characteristics in this context could be: type of holiday (sea-sun-sand vs. ski), travel motive (adventure vs. culture), travel season (summer vs. winter), length of trip (weekend vs. week/s), travel companion (single vs. family), or mode of transport (airplane vs. car). A tourist who is deciding on a weekend shopping trip will probably consider a different number of alternatives as when deciding on a three week adventure backpacking trip. Nevertheless, studies on set theory have not yet concentrated on the question if and how tourists' relevant sets differ depending on the characteristics of a holiday and further research needs to integrate this variety of DC sets.

References

- Amemiya, E. C. (1963). Measurement of economic differentiation. *Journal of Regional Science*, 5(1), 85–87.
- Ankomah, P. K., Crompton, J. L., & Baker, D. (1996). Influence of cognitive distance in vacation choice. *Annals of Tourism Research*, 23(1), 138–150.
- Botha, C., Crompton, J. L., & Kim, S.-S. (1999). Developing a revised competitive position for Sun/Lost City, South Africa. *Journal of Travel Research*, 37(4), 341–352.
- Cohen, E. (1972). Towards a sociology of international tourism. *Social Research*, 39(1), 164–182.
- Crompton, J. L. (1992). Structure of vacation destination choice sets. *Annals of Tourism Research*, 19(3), 420–434.
- Crompton, J. L., & Ankomah, P. K. (1993). Choice set propositions in destination decisions. *Annals of Tourism Research*, 20(3), 461–476.
- Crompton, J. L., & Kim, S.-S. (2001). The influence of cognitive distance in vacation choice. *Annals of Tourism Research*, 28(2), 512–515.
- Decrop, A. (2006). *Vacation decision making*. Wallingford: CABI Pub..
- Decrop, A. (2010). Destination choice sets: An inductive longitudinal approach. *Annals of Tourism Research*, 37(1), 93–115.
- Decrop, A., & Snelders, D. (2005). A grounded typology of vacation decision-making. *Tourism Management*, 26(2), 121–132.
- Decrop, A., & Zidda, P. (2006). Typology of vacation decision-making modes. *Tourism Analysis*, 11(3), 189–197.
- Deutsch-Burgner, K., Ravualaparthi, S., & Goulias, K. (2014). Place happiness: Its constituents and the influence of emotions and subjective importance on activity type and destination choice. *Transportation*, 41(6), 1323–1340.
- Fesenmaier, D. R., & Jeng, J.-M. (2000). Assessing the structure in the pleasure trip planning process. *Tourism Analysis*, 5(1), 13–27.
- Gilbert, D., & Hudson, S. (2000). Tourism demand constraints: A skiing participation. *Annals of Tourism Research*, 27(4), 906–925.
- Hofstede, G., Hofstede, G., & Minkov, M. (2010). *Cultures and organizations: Software of the mind*. New York: McGraw-Hill.

- Howard, J. A., & Sheth, J. N. (1969). *The theory of buyer behavior*. New York: Wiley.
- IEP (Institute for Economics and Peace). (2013). *Global peace index*. Retrieved December 16, 2014 from IEP website: <<http://www.visionofhumanity.org/#page/indexes/global-peace-index/2013>>.
- Jang, H., Lee, S., Lee, S.-W., & Hong, S.-K. (2007). Expanding the individual choice-sets model to couples' honeymoon destination selection process. *Tourism Management*, 28(5), 1299–1314.
- Jeng, J., & Fesenmaier, D. R. (2002). Conceptualizing the travel decision-making hierarchy: A review of recent developments. *Tourism Analysis*, 7(1), 15–32.
- Lepp, A., & Gibson, H. (2008). Sensation seeking and tourism: Tourist role, perception of risk and destination choice. *Tourism Management*, 29(4), 740–750.
- Litvin, S. W., Goldsmith, R. E., & Pan, B. (2008). Electronic word-of-mouth in hospitality and tourism management. *Tourism Management*, 29(3), 458–468.
- Lohmann, M., & Aderhold, P. (2009). *Urlaubsreisetrends 2020. Die RA-Trendstudie – Entwicklung der touristischen Nachfrage der Deutschen* [Leisure travel trends 2020. Travel analysis trend study – Development of the German tourism demand]. Kiel: F.U.R.
- Mansfeld, Y. (1992). From motivation to actual travel. *Annals of Tourism Research*, 19(3), 399–419.
- Marcussen, C. H. (2011). Understanding destination choices of German travelers. *Tourism Analysis*, 16(6), 649–662.
- Martin, D., & Woodside, A. G. (2012). Structure and process modeling of seemingly unstructured leisure-travel decisions and behaviour. *International Journal of Contemporary Management*, 24(6), 855–872.
- Mathieson, A., & Wall, G. (1982). *Tourism: Economic, physical and social impacts*. Harlow: Longman Scientific & Technical.
- Money, R. B., & Crofts, J. C. (2003). The effect of uncertainty avoidance on information search, planning, and purchases of international travel vacations. *Tourism Management*, 24(2), 191–202.
- Moutinho, L. (1987). Consumer behaviour in tourism. *European Journal of Marketing*, 21(10), 5–44.
- Narayana, C., & Markin, R. (1975). Consumer behavior and product performance: An alternative conceptualization. *Journal of Marketing*, 39(1), 1–6.
- Nyaupane, G. P., & Andereck, K. L. (2007). Understanding travel constraints: Application and extension of a leisure constraint model. *Journal of Travel Research*, 46(4), 433–439.
- Papatheodorou, A. (2001). Why people travel to different places. *Annals of Tourism Research*, 28(1), 164–179.
- Park, J.-Y., & Jang, S. (2013). Confused by too many choices? Choice overload in tourism. *Tourism Management*, 35, 1–12.
- Perdue, H., & Meng, F. (2006). Understanding choice and rejection in destination consideration sets. *Tourism Analysis*, 11(6), 337–348.
- Plog, S. C. (1974). Why destination areas rise and fall in popularity. *Cornell Hotel and Restaurant Administration Quarterly*, 14(4), 55–58.
- Prentice, R. (2006). Evocation and experiential seduction: Updating choice-sets modelling. *Tourism Management*, 27(6), 1153–1170.
- Reisinger, Y., & Mavondo, F. (2005). Travel anxiety and intentions to travel internationally: Implications of travel risk perception. *Journal of Travel Research*, 43(3), 212–225.
- Ryan, C. (1998). The travel career ladder. An appraisal. *Annals of Tourism Research*, 25(4), 936–957.
- Seddighi, H. R., & Theodorou, A. L. (2002). A model of tourism destination choice: A theoretical and empirical analysis. *Tourism Management*, 23(5), 475–487.
- Sirakaya, E., & Woodside, A. G. (2005). Building and testing theories of decision making by travellers. *Tourism Management*, 26(6), 815–832.
- Smallman, C., & Moore, K. (2010). Process studies of tourists' decision-making. *Annals of Tourism Research*, 37(2), 397–422.
- Sönmez, S., & Graefe, A. (1998). Determining future travel behavior from past travel experience and perceptions of risk and safety. *Journal of Travel Research*, 37(2), 171–177.
- Sousa, C. M. P., & Bradley, F. (2006). Cultural distance and psychic distance: Two peas in a pod? *Journal of International Marketing*, 14(1), 49–70.
- Spiggle, S., & Sewall, M. A. (1987). A choice sets model of retail selection. *Journal of Marketing*, 51(2), 97–111.
- Um, S., & Crompton, J. L. (1990). Attitude determinants in tourism destination choice. *Annals of Tourism Research*, 17(3), 432–448.
- Um, S., & Crompton, J. L. (1992). The roles of perceived inhibitors and facilitators in pleasure travel destination decisions. *Journal of Travel Research*, 30(3), 18–25.
- UNDP (United Nations Development Programme). (2013). *Human development report 2013. The rise of the south: Human progress in a diverse world*. Kanata: Gilmore Printing Services Inc..
- UNWTO (World Tourism Organization). (2013a). *Tourism highlights. 2013 edition*. Madrid: UNWTO.
- UNWTO (World Tourism Organization). (2013b). *Yearbook of tourism statistics. 2013 edition. Data 2007–2011*. Madrid: UNWTO.
- van Raaij, W., & Francken, D. A. (1984). Vacation decisions, activities, and satisfactions. *Annals of Tourism Research*, 11(1), 101–112.
- Williams, A. M., & Baláž, V. (2014). Tourism risk and uncertainty: Theoretical reflections. *Journal of Travel Research*, 24, 1–17.
- Woodside, A. G., & Lysonski, S. (1989). A general model of traveler destination choice. *Journal of Travel Research*, 27(4), 8–14.
- Woodside, A. G., & Sherrell, D. (1977). Traveler evoked, inept, and inert sets of vacation destinations. *Journal of Travel Research*, 16(1), 14–18.